

Initiating Search

November 10, 2025, 4:16 PM

• Search:

CAS Lexicon Search:

Fluorination - Preferred Concept

Search Tasks

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Returned Reference Results from CAS Lexicon (21,000)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,001)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

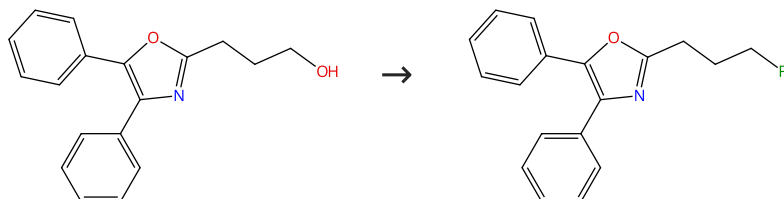
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 Reactions (50)[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (3)

31-048-CAS-18660808

Steps: 1 Yield: 100%

Deoxyfluorination with Sulfonyl Fluorides: Navigating Reaction Space with Machine Learning

By: Nielsen, Matthew K.; et al

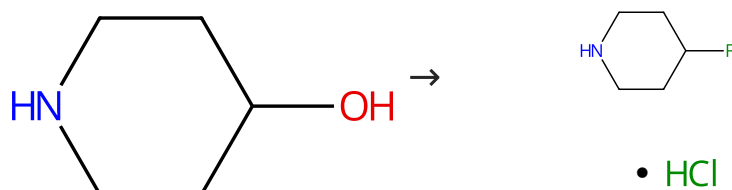
Journal of the American Chemical Society (2018), 140(15), 5004-5008.

1.1 Reagents: 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine, 4-(Trifluoromethyl)benzenesulfonyl fluoride
Solvents: Tetrahydrofuran; 48 h, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (89) Suppliers (69)

31-048-CAS-12323230

Steps: 1 Yield: 100%

Fluorodehydroxylation, a novel method for synthesis of fluoro amines and fluoroamino acids

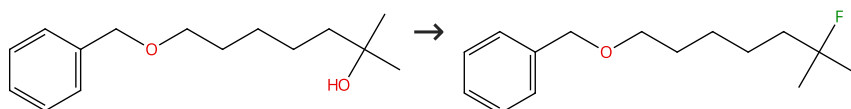
By: Kollonitsch, J.; et al

Journal of Organic Chemistry (1979), 44(5), 771-7.

1.1 Reagents: Hydrofluoric acid

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-37847883

Steps: 1 Yield: 100%

Synthesis of Tertiary Fluorides Through an Acid-Mediated Deoxyfluorination of Tertiary Alcohols

By: Bertrand, Xavier; et al

Journal of Organic Chemistry (2023), 88(20), 14527-14539.

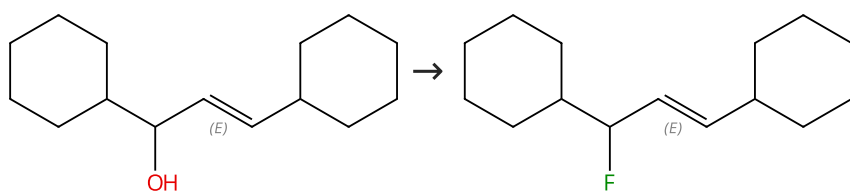
1.1 Reagents: Methanesulfonic acid, Potassium bifluoride
Solvents: Dichloromethane; 0 °C; 1 h, 0 °C

1.2 Reagents: Sodium bicarbonate
Solvents: Water

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

Supplier (1)

Suppliers (5)

31-048-CAS-5311054

Steps: 1 Yield: 100%

Interactions of cationic palladium(II)- and platinum(II)- η^3 -allyl complexes with fluoride: is asymmetric allylic fluorination a viable reaction?

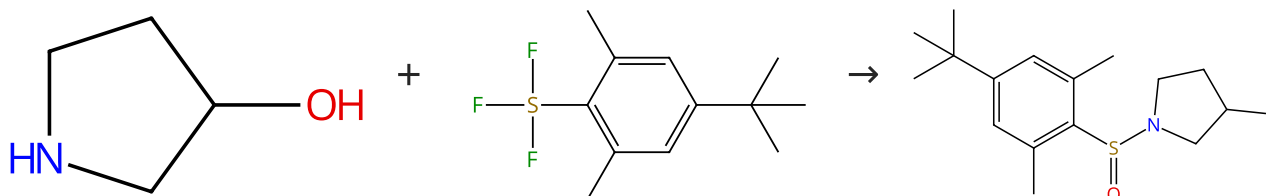
By: Hintermann, Lukas; et al

European Journal of Inorganic Chemistry (2006), (7), 1397-1412.

- 1.1 **Reagents:** (Diethylamino)sulfur trifluoride
Solvents: Dichloromethane; -78 °C; -78 °C → 0 °C; 10 min, 0 °C
- 1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

Suppliers (56)

Supplier (1)

31-048-CAS-3505131

Steps: 1 Yield: 100%

Discovery of 4-tert-Butyl-2,6-dimethylphenylsulfur Trifluoride as a Deoxofluorinating Agent with High Thermal Stability as Well as Unusual Resistance to Aqueous Hydrolysis, and Its Diverse Fluorination Capabilities Including Deoxofluoro-Arylsulfonylation with High Stereoselectivity

By: Umemoto, Teruo; et al

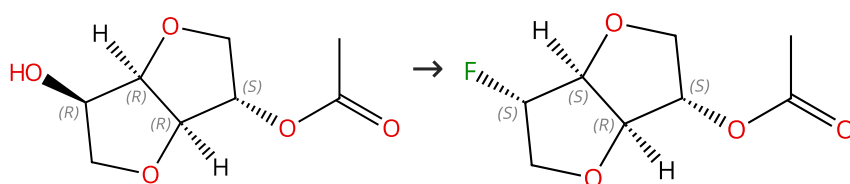
Journal of the American Chemical Society (2010), 132(51), 18199-18205.

- 1.1 **Reagents:** Pyridine, hydrofluoride (1:1)
Solvents: Dichloromethane; rt
- 1.2 **Solvents:** Dichloromethane; rt; 4 h, rt
- 1.3 **Reagents:** Triethylamine; rt; 2 h, rt

Experimental Protocols

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%

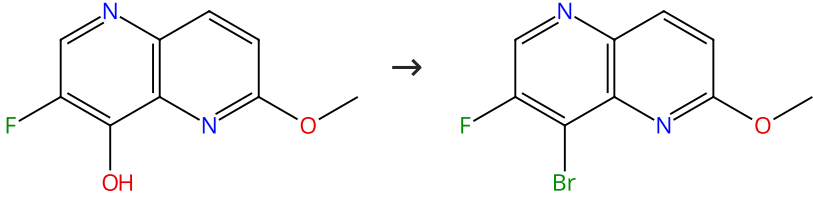


Absolute stereochemistry shown

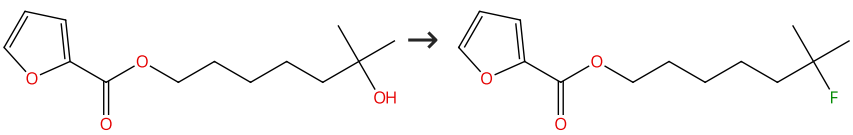
Absolute stereochemistry shown

Suppliers (42)

31-048-CAS-18660814 Steps: 1 Yield: 100% 1.1 Reagents: 1,1,2,2,3,3,4,4,4-Nonafluoro-1-butanesulfonyl fluoride, 2-Methyl-<i>N</i>-(tri-1-pyrrolidinylphosphoranylidene)-2-propanamine Solvents: Tetrahydrofuran; 48 h, rt Experimental Protocols	Deoxyfluorination with Sulfonyl Fluorides: Navigating Reaction Space with Machine Learning By: Nielsen, Matthew K.; et al Journal of the American Chemical Society (2018), 140(15), 5004-5008.
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Scheme 7 (1 Reaction) Steps: 1 Yield: 100%  🛒 Suppliers (34) 🛒 Suppliers (59)
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31-048-CAS-14622107 Steps: 1 Yield: 100% 1.1 Reagents: Phosphorus tribromide Solvents: Dimethylformamide; 50 °C; 1 h, 70 °C Experimental Protocols	A One-Pot Diazotation-Fluorodediazoniation Reaction and Fluorine Gas for the Production of Fluoronaphthyridines By: Abele, Stefan; et al Organic Process Research & Development (2014), 18(8), 993-1001.
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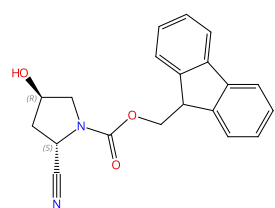
Scheme 8 (2 Reactions) Steps: 1 Yield: 98-100% 

31-614-CAS-41687069 Steps: 1 Yield: 100% 1.1 Reagents: Sulfuric acid, Potassium fluoride Solvents: Dichloromethane; 0 °C; 1 h, 0 °C 1.2 Reagents: Sodium bicarbonate Solvents: Water Experimental Protocols	Improved conditions for the synthesis of tertiary fluorides using a KF/H₂SO₄ combination By: Bertrand, Xavier; et al Journal of Fluorine Chemistry (2024), 279, 110338.
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31-614-CAS-37847899 Steps: 1 Yield: 98% 1.1 Reagents: Methanesulfonic acid, Potassium bifluoride Solvents: Dichloromethane; 0 °C; 1 h, 0 °C 1.2 Reagents: Sodium bicarbonate Solvents: Water Experimental Protocols	Synthesis of Tertiary Fluorides Through an Acid-Mediated Deoxyfluorination of Tertiary Alcohols By: Bertrand, Xavier; et al Journal of Organic Chemistry (2023), 88(20), 14527-14539.
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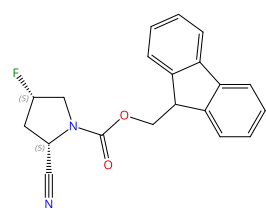
Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Supplier (1)



Absolute stereochemistry shown

Suppliers (9)

31-048-CAS-10868328

Steps: 1 Yield: 100%

1.1 **Reagents:** (7-4)-[4-(1,1-Dimethylethyl)-2,6-dimethylphenyl] trifluorosulfur

Solvents: Dichloromethane; 0 °C; 1 h, 0 °C; 0 °C → rt; 60 h, rt

Experimental Protocols

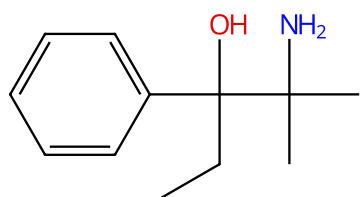
Discovery of 4-tert-Butyl-2,6-dimethylphenylsulfur Trifluoride as a Deoxofluorinating Agent with High Thermal Stability as Well as Unusual Resistance to Aqueous Hydrolysis, and Its Diverse Fluorination Capabilities Including Deoxofluoro-Arylsulfinylation with High Stereoselectivity

By: Umemoto, Teruo; et al

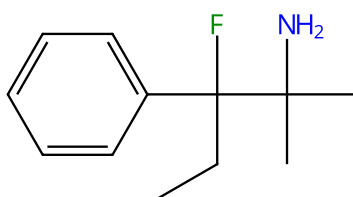
Journal of the American Chemical Society (2010), 132(51), 18199-18205.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)



Suppliers (2)

31-048-CAS-2237209

Steps: 1 Yield: 100%

No Data Available

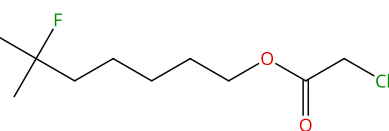
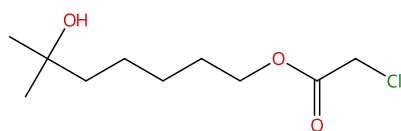
Fluorination of amino alcohols and hydroxyaziridines by Olah's reagent

By: Alvernhe, Gerard; et al

Journal of Chemical Research, Synopses (1983), (10), 246-7.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-37847901

Steps: 1 Yield: 100%

1.1 **Reagents:** Methanesulfonic acid, Potassium bifluoride

Solvents: Dichloromethane; 0 °C; 1 h, 0 °C

1.2 **Reagents:** Sodium bicarbonate

Solvents: Water

Experimental Protocols

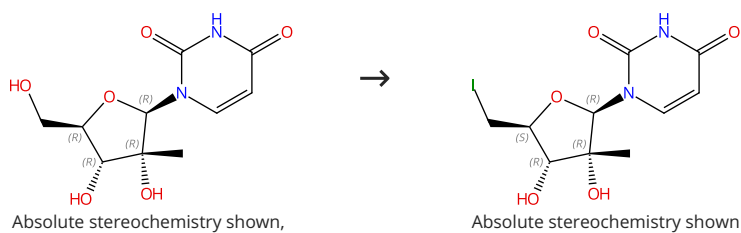
Synthesis of Tertiary Fluorides Through an Acid-Mediated Deoxyfluorination of Tertiary Alcohols

By: Bertrand, Xavier; et al

Journal of Organic Chemistry (2023), 88(20), 14527-14539.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (66)

Suppliers (6)

31-048-CAS-19310529

Steps: 1 Yield: 100%

Structure-activity relationship of uridine-based nucleoside phosphoramidate prodrugs for inhibition of dengue virus RN A-dependent RNA polymerase

1.1 **Reagents:** Pyridine, Imidazole, Triphenyl phosphine, Iodine
Solvents: Tetrahydrofuran; 1.5 h, rt

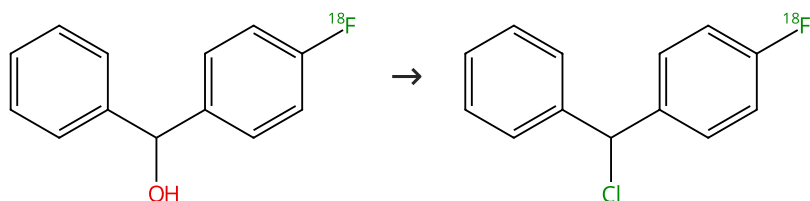
Experimental Protocols

By: Wang, Gang; et al

Bioorganic & Medicinal Chemistry Letters (2018), 28(13), 2324-2327.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



31-048-CAS-14719056

Steps: 1 Yield: 100%

Aryltrimethylammonium trifluoromethanesulfonates as precursors to aryl [¹⁸F]fluorides: improved synthesis of [¹⁸F]G BR-13119

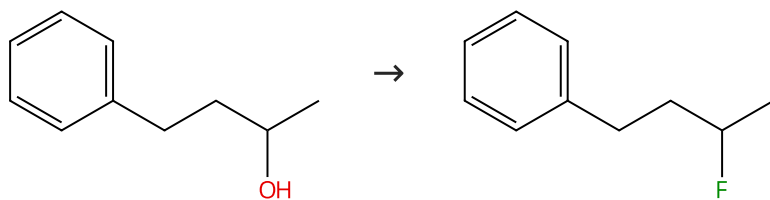
1.1 **Reagents:** Thionyl chloride

By: Haka, Michael S.; et al

Journal of Labelled Compounds and Radiopharmaceuticals (1989), 27(7), 823-33.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (63)

Suppliers (4)

31-048-CAS-189777

Steps: 1 Yield: 100%

Fluorine Transfer to Alkyl Radicals

1.1 **Reagents:** (7-4)-Trifluoro(*N*-methylmethanaminato)sulfur
Solvents: Dichloromethane; 30 min, 0 °C; 0 °C → rt

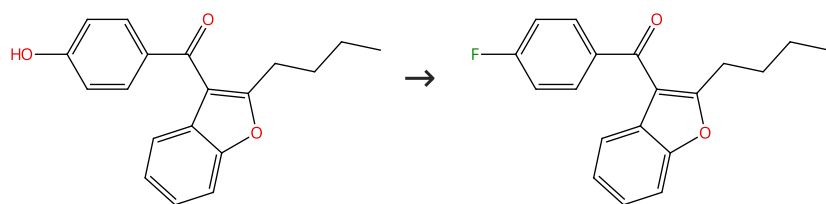
By: Rueda-Becerril, Montserrat; et al

Experimental Protocols

Journal of the American Chemical Society (2012), 134(9), 4026-4029.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (76)

31-614-CAS-36549707

Steps: 1 Yield: 100%

Deoxyfluorination of Electron-Deficient Phenols

- 1.1 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene, 1 *H*-imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-2-chloro-, (dihydrogen trifluoride) (1:1)
Solvents: Toluene; rt; 16 h, 95 °C

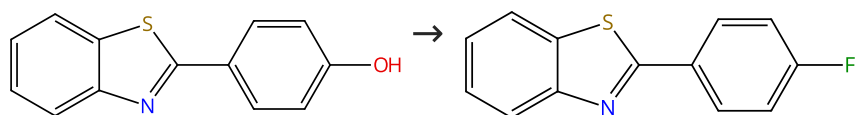
By: Jelen, Jan; et al

Organic Letters (2023), 25(20), 3649-3653.

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (54)

Suppliers (43)

31-048-CAS-16648019

Steps: 1 Yield: 99%

Concerted nucleophilic aromatic substitution with $^{19}\text{F}^-$ and $^{18}\text{F}^-$

- 1.1 **Reagents:** Cesium fluoride, 1 *H*-imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-2-chloro-, chloride (1:1), mixt. with cesium fluoride
Solvents: Toluene; 30 min, 23 °C; 10 h, 110 °C

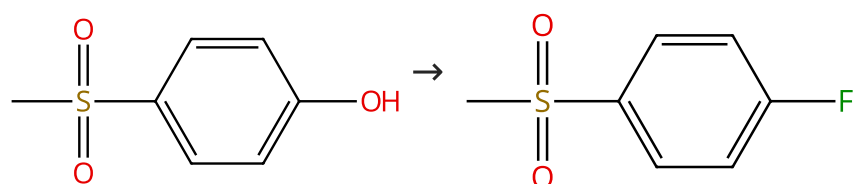
By: Neumann, Constanze N.; et al

Nature (London, United Kingdom) (2016), 534(7607), 369-373.

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (77)

Suppliers (74)

31-048-CAS-3899942

Steps: 1 Yield: 99%

PhenoFluorMix: Practical Chemoselective Deoxyfluorination of Phenols

- 1.1 **Reagents:** 1 *H*-imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-2-chloro-, chloride (1:1), mixt. with cesium fluoride
Solvents: Toluene; 30 min, 23 °C; 24 h, 110 °C

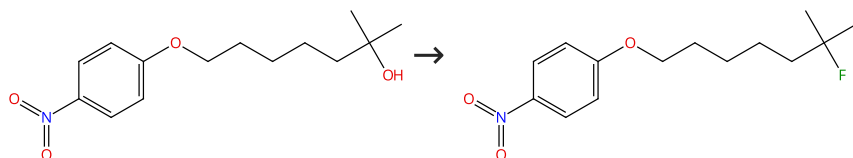
By: Fujimoto, Teppei; et al

Organic Letters (2015), 17(3), 544-547.

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-37847894

Steps: 1 Yield: 99%

1.1 **Reagents:** Methanesulfonic acid, Potassium bifluoride
Solvents: Dichloromethane; 0 °C; 1 h, 0 °C

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

Experimental Protocols

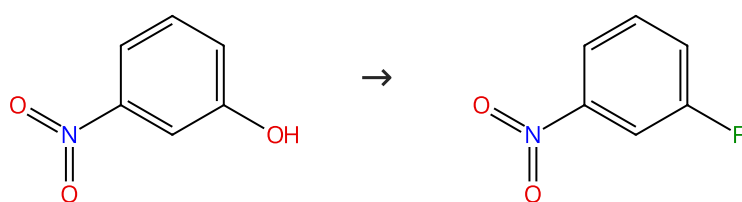
Synthesis of Tertiary Fluorides Through an Acid-Mediated Deoxyfluorination of Tertiary Alcohols

By: Bertrand, Xavier; et al

Journal of Organic Chemistry (2023), 88(20), 14527-14539.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (87)

Suppliers (65)

31-614-CAS-36549697

Steps: 1 Yield: 99%

1.1 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene, 1-*H*-imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-2-chloro-, (dihydrogen trifluoride) (1:1)
Solvents: Toluene; rt; 16 h, 95 °C

Experimental Protocols

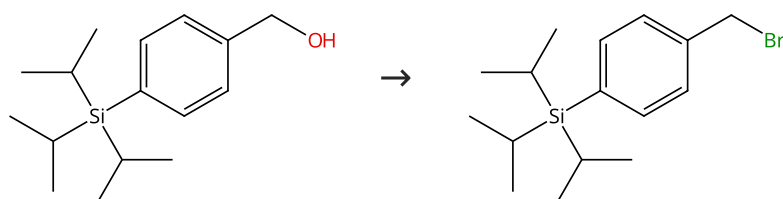
Deoxyfluorination of Electron-Deficient Phenols

By: Jelen, Jan; et al

Organic Letters (2023), 25(20), 3649-3653.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (3)

31-048-CAS-21935841

Steps: 1 Yield: 99%

1.1 **Reagents:** Carbon tetrabromide, Triphenylphosphine
Solvents: Dichloromethane; 2.5 h, 0 °C

Experimental Protocols

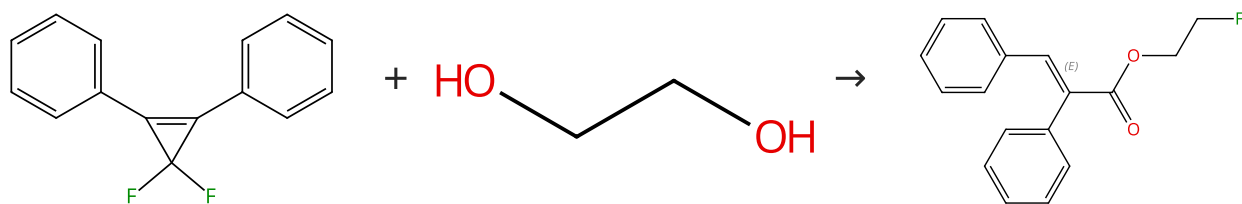
Catalytic Asymmetric Fluorination of Copper Carbene Complexes: Preparative Advances and a Mechanistic Rationale

By: Buchsteiner, Michael; et al

Chemistry - A European Journal (2020), 26(11), 2509-2515.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (8)

Suppliers (166)

Double bond geometry shown

31-048-CAS-17132228

Steps: 1 Yield: 99%

Deoxyfluorination of alcohols with 3,3-difluoro-1,2-diarylcyclopropenes

1.1 Solvents: 1,2-Dichloroethane; 8 h, 80 °C

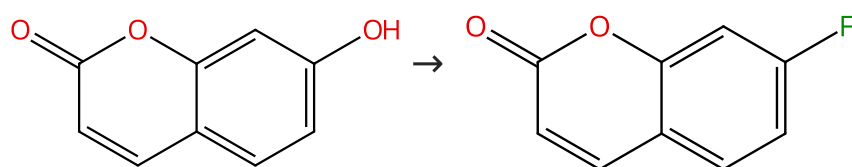
By: Li, Lingchun; et al

Experimental Protocols

Nature Communications (2016), 7, 13320pp..

Scheme 22 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (113)

Suppliers (39)

31-614-CAS-36549700

Steps: 1 Yield: 99%

Deoxyfluorination of Electron-Deficient Phenols

1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene, 1*H*-imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-2-chloro-, (dihydrogen trifluoride) (1:1)
Solvents: Toluene; rt; 12 h, 85 °C

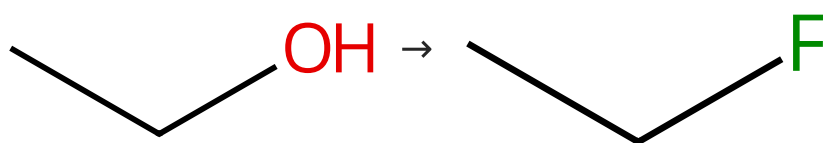
By: Jelen, Jan; et al

Organic Letters (2023), 25(20), 3649-3653.

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (620)

Suppliers (12)

31-048-CAS-5271511

Steps: 1 Yield: 99%

Uranium fluoride chemistry. Part 1. The gas phase reaction of uranium hexafluoride with alcohols

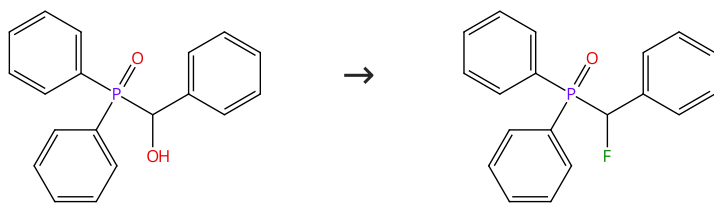
1.1 Reagents: Uranium hexafluoride

By: Schnautz, Norman G.; et al

South African Journal of Chemistry (1992), 45(2-3), 59-62.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (6)

Suppliers (2)

31-048-CAS-5210682

Steps: 1 Yield: 99%

Synthesis of α - and γ -fluoroalkylphosphonates

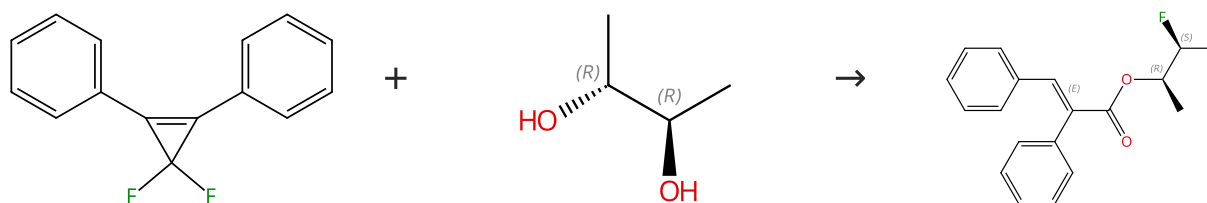
1.1 **Reagents:** (Diethylamino)sulfur trifluoride
Solvents: Dichloromethane

By: Blackburn, G. Michael; et al

Journal of the Chemical Society, Perkin Transactions 1:
 Organic and Bio-Organic Chemistry (1972-1999) (1986), (6),
 913-17.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (8)

Absolute stereochemistry shown,
 Rotation (-)

Absolute stereochemistry shown
 Double bond geometry shown

Suppliers (80)

31-048-CAS-17132236

Steps: 1 Yield: 99%

Deoxyfluorination of alcohols with 3,3-difluoro-1,2-diarylcyclopropenes

1.1 **Solvents:** 1,2-Dichloroethane; 8 h, 80 °C

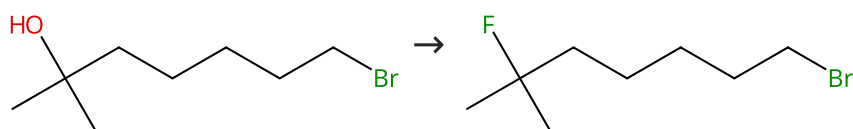
Experimental Protocols

By: Li, Lingchun; et al

Nature Communications (2016), 7, 13320pp..

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (33)

Suppliers (5)

31-614-CAS-37847897

Steps: 1 Yield: 99%

Synthesis of Tertiary Fluorides Through an Acid-Mediated Deoxyfluorination of Tertiary Alcohols

1.1 **Reagents:** Methanesulfonic acid, Potassium bifluoride
Solvents: Dichloromethane; 0 °C; 1 h, 0 °C

By: Bertrand, Xavier; et al

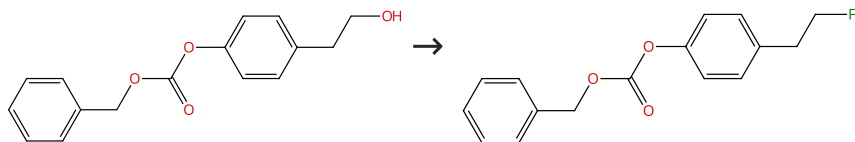
1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

Journal of Organic Chemistry (2023), 88(20), 14527-14539.

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-38381882

Steps: 1 Yield: 99%

Sulfur Tetrafluoride (SF₄) as a Deoxyfluorination Reagent for Organic Synthesis in Continuous Flow Mode

By: Polterauer, Dominik; et al

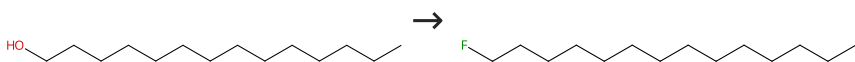
Organic Process Research & Development (2023), 27(12), 2385-2392.

1.1 Reagents: Triethylamine, Sulfur tetrafluoride
Solvents: Ethyl acetate; 5 min, 50 °C1.2 Reagents: Sodium hydroxide
Solvents: Water

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (87)

Suppliers (15)

31-614-CAS-33631175

Steps: 1 Yield: 99%

Synthesis of Highly Reactive Sulfone Iminium Fluorides and Their Use in Deoxyfluorination and Sulfur Fluoride Exchange Chemistry

By: Vogel, James A.; et al

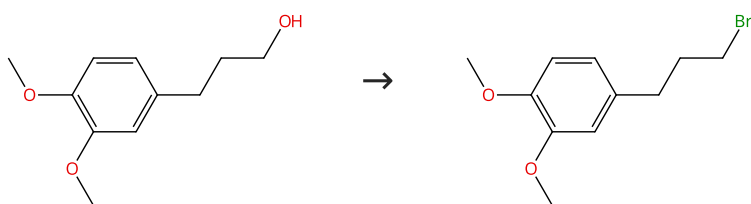
Organic Letters (2022), 24(32), 5962-5966.

1.1 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene, 2815253-79-1
Solvents: Dichloromethane; 60 s, rt

Experimental Protocols

Scheme 29 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (52)

Suppliers (41)

31-048-CAS-23731130

Steps: 1 Yield: 99%

Inter- and intra-molecular organocatalysis of S_N2 fluorination by crown ether: kinetics and quantum chemical analysis

By: Oh, Young-Ho; et al

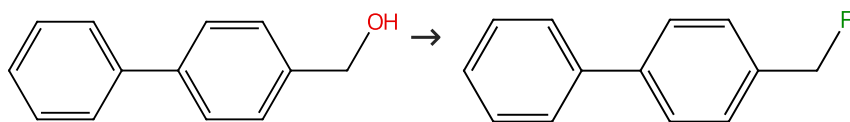
Molecules (2021), 26(10), 2947.

1.1 Reagents: N-Bromosuccinimide, Triphenylphosphine
Solvents: Dichloromethane; 1 h, 0 °C1.2 Reagents: Sodium bicarbonate
Solvents: Water

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (85)

Suppliers (2)

31-048-CAS-18660803

Steps: 1 Yield: 99%

Deoxyfluorination with Sulfonyl Fluorides: Navigating Reaction Space with Machine Learning

By: Nielsen, Matthew K.; et al

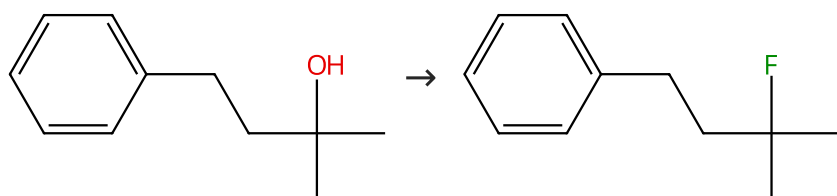
Journal of the American Chemical Society (2018), 140(15), 5004-5008.

1.1 **Reagents:** 4-(Trifluoromethyl)benzenesulfonyl fluoride, 2-Methyl-*N*-(tri-1-pyrrolidinylphosphoranylidene)-2-propan amine
Solvents: Tetrahydrofuran; 48 h, rt

Experimental Protocols

Scheme 31 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (66)

Suppliers (2)

31-048-CAS-22532913

Steps: 1 Yield: 99%

Dehydroxylation Fluorination of Tertiary Alcohols

By: Zhang, Wei; et al

Organic Letters (2020), 22(16), 6642-6646.

1.1 **Reagents:** 1,2-Diiodoethane, 1,2-Bis(diphenylphosphino)ethane, Zinc bromide, Selectfluor
Solvents: Acetonitrile; 15 min, rt

1.2 **Solvents:** Water

31-048-CAS-20833999

Steps: 1 Yield: 99%

Activation of primary and secondary benzylic and tertiary alkyl (sp³)C-F bonds inside a self-assembled molecular container

By: Koster, Jesper M.; et al

Frontiers in Chemistry (Lausanne, Switzerland) (2018), 6, 639.

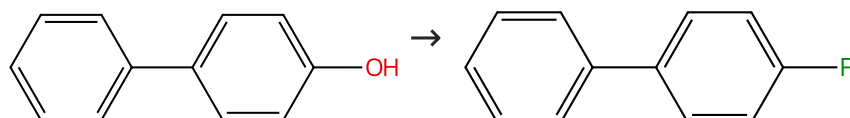
1.1 **Reagents:** (Diethylamino)sulfur trifluoride
Solvents: Dichloromethane; -78 °C; 1 h, -78 °C

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 99%



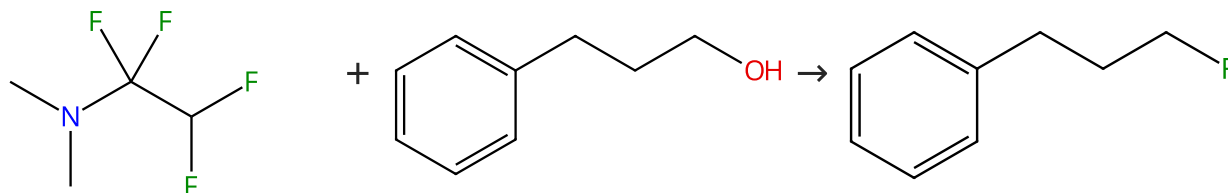
Suppliers (105)

Suppliers (76)

31-048-CAS-9434614	Steps: 1 Yield: 99%	PhenoFluorMix: Practical Chemoselective Deoxyfluorination of Phenols
1.1 Reagents: Cesium fluoride, 1 <i>H</i> -imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-2-chloro-, chloride (1:1) Solvents: Ethyl acetate; 30 min, 23 °C; 24 h, 100 °C		By: Fujimoto, Teppei; et al Organic Letters (2015), 17(3), 544-547.
Experimental Protocols		

Scheme 33 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (25)

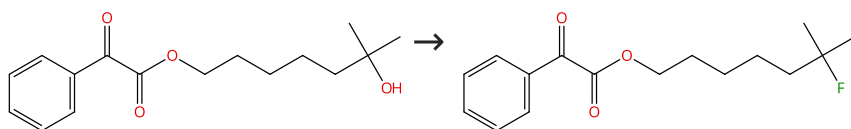
Suppliers (87)

Suppliers (10)

31-048-CAS-518147	Steps: 1 Yield: 98%	1,1,2,2-Tetrafluoroethyl-N,N-dimethylamine: a new selective fluorinating agent
1.1 Solvents: Chloroform 1.2 Reagents: Hydrochloric acid Solvents: Water		By: Petrov, V. A.; et al Journal of Fluorine Chemistry (2001), 109(1), 25-31.

Scheme 34 (1 Reaction)

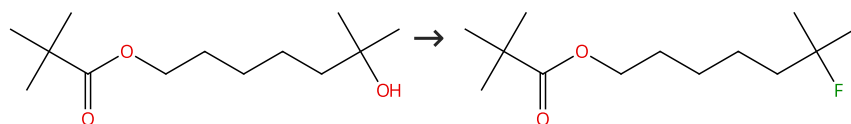
Steps: 1 Yield: 98%



31-614-CAS-37847890	Steps: 1 Yield: 98%	Synthesis of Tertiary Fluorides Through an Acid-Mediated Deoxyfluorination of Tertiary Alcohols
1.1 Reagents: Methanesulfonic acid, Potassium bifluoride Solvents: Dichloromethane; 0 °C; 1 h, 0 °C 1.2 Reagents: Sodium bicarbonate Solvents: Water		By: Bertrand, Xavier; et al Journal of Organic Chemistry (2023), 88(20), 14527-14539.
Experimental Protocols		

Scheme 35 (1 Reaction)

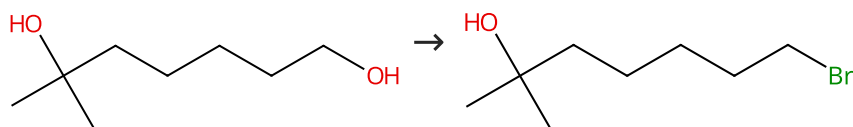
Steps: 1 Yield: 98%



31-614-CAS-37847887	Steps: 1 Yield: 98%	Synthesis of Tertiary Fluorides Through an Acid-Mediated Deoxyfluorination of Tertiary Alcohols
1.1 Reagents: Methanesulfonic acid, Potassium bifluoride Solvents: Dichloromethane; 0 °C; 1 h, 0 °C 1.2 Reagents: Sodium bicarbonate Solvents: Water		By: Bertrand, Xavier; et al Journal of Organic Chemistry (2023), 88(20), 14527-14539.
Experimental Protocols		

Scheme 36 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (22)

Suppliers (33)

31-048-CAS-22532908

Steps: 1 Yield: 98%

Dehydroxylative Fluorination of Tertiary Alcohols

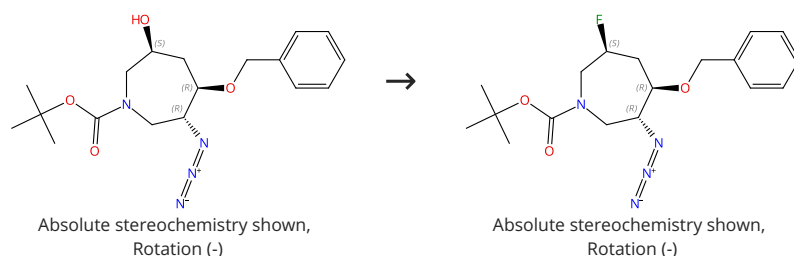
1.1 Reagents: Carbon tetrabromide, Triphenylphosphine; 0 °C → rt; overnight, rt

By: Zhang, Wei; et al

Organic Letters (2020), 22(16), 6642-6646.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 98%



31-048-CAS-13547435

Steps: 1 Yield: 98%

Conformational regulation of substituted azepanes through selective monofluorination

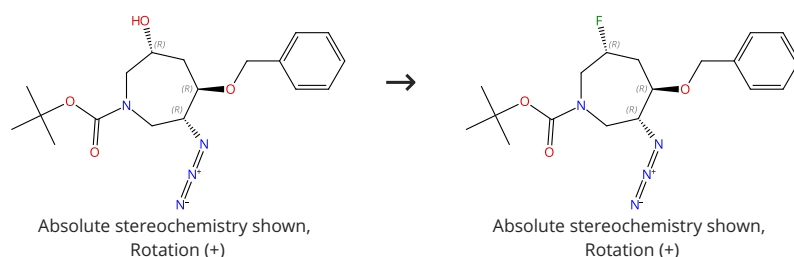
1.1 Reagents: Deoxo-Fluor
Solvents: Dichloromethane; 1.5 h, 25 °C

By: Patel, Alpesh Ramanlal; et al

Organic & Biomolecular Chemistry (2013), 11(23), 3781-3785.

Scheme 38 (1 Reaction)

Steps: 1 Yield: 98%



31-048-CAS-471927

Steps: 1 Yield: 98%

Conformational regulation of substituted azepanes through selective monofluorination

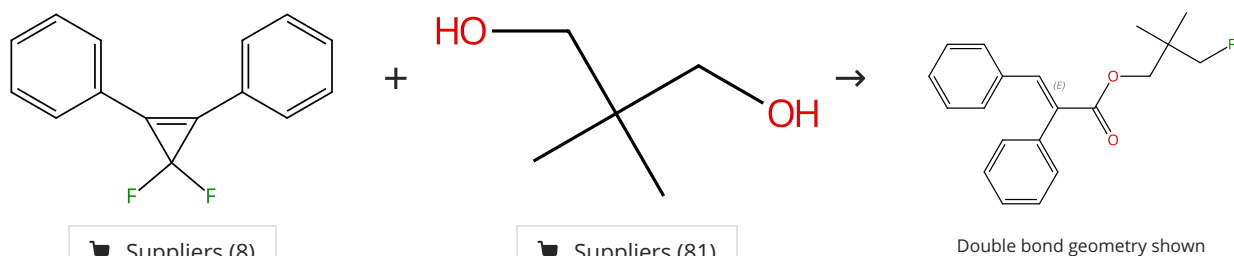
1.1 Reagents: Deoxo-Fluor
Solvents: Dichloromethane; 1.5 h, 25 °C

By: Patel, Alpesh Ramanlal; et al

Organic & Biomolecular Chemistry (2013), 11(23), 3781-3785.

Scheme 39 (1 Reaction)

Steps: 1 Yield: 98%



31-048-CAS-17132239

Steps: 1 Yield: 98%

Deoxyfluorination of alcohols with 3,3-difluoro-1,2-diarylcyclopropenes

1.1 Solvents: 1,2-Dichloroethane; 8 h, 80 °C

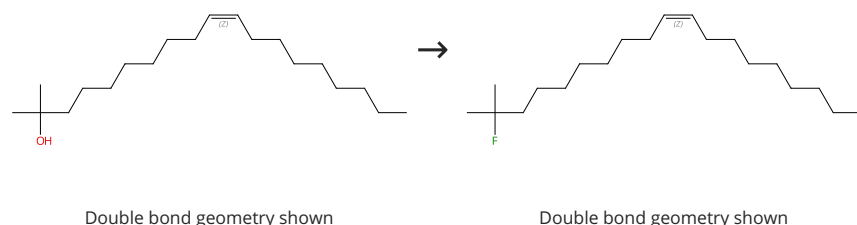
Experimental Protocols

By: Li, Lingchun; et al

Nature Communications (2016), 7, 13320pp..

Scheme 40 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-37847891

Steps: 1 Yield: 98%

Synthesis of Tertiary Fluorides Through an Acid-Mediated Deoxyfluorination of Tertiary Alcohols

1.1 Reagents: Methanesulfonic acid, Potassium bifluoride

Solvents: Dichloromethane; 0 °C; 1 h, 0 °C

1.2 Reagents: Sodium bicarbonate

Solvents: Water

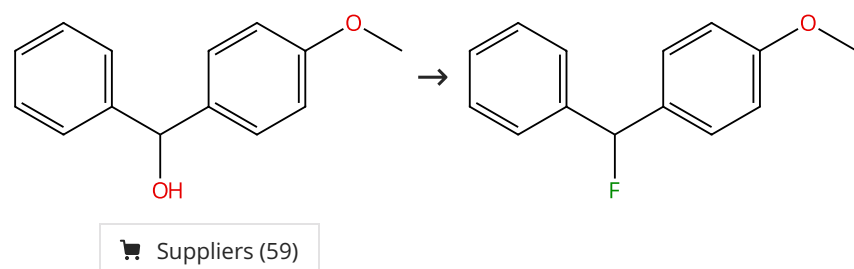
Experimental Protocols

By: Bertrand, Xavier; et al

Journal of Organic Chemistry (2023), 88(20), 14527-14539.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-41865171

Steps: 1 Yield: 98%

Visible light-induced photocatalytic deoxyfluorination of benzyl alcohol using SF₆ as a fluorinating reagent

1.1 Reagents: Sulfur hexafluoride, Diisopropylethylamine

Catalysts: 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile

Solvents: 1,2-Dichloroethane; 20 h, 50 °C

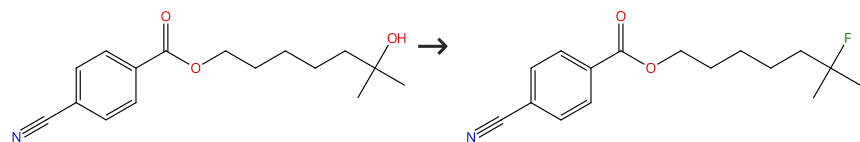
Experimental Protocols

By: Zhang, Yi-Fan; et al

Green Chemistry (2024), 26(19), 10324-10329.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-37847889

Steps: 1 Yield: 98%

1.1 Reagents: Methanesulfonic acid, Potassium bifluoride
Solvents: Dichloromethane; 0 °C; 1 h, 0 °C

1.2 Reagents: Sodium bicarbonate
Solvents: Water

Experimental Protocols

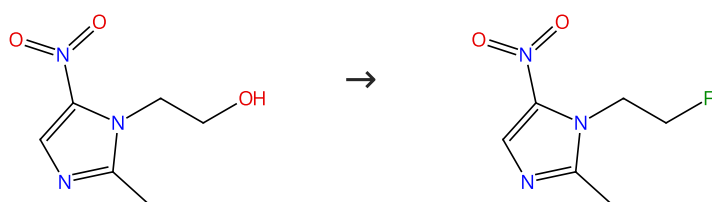
Synthesis of Tertiary Fluorides Through an Acid-Mediated Deoxyfluorination of Tertiary Alcohols

By: Bertrand, Xavier; et al

Journal of Organic Chemistry (2023), 88(20), 14527-14539.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (129)

Suppliers (4)

31-048-CAS-16375868

Steps: 1 Yield: 98%

1.1 Reagents: Potassium fluoride, 1*H*-imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-2-fluoro-, tetrafluoroborate(1-) (1:1)
Solvents: 1,4-Dioxane; 17 h, 40 °C

Experimental Protocols

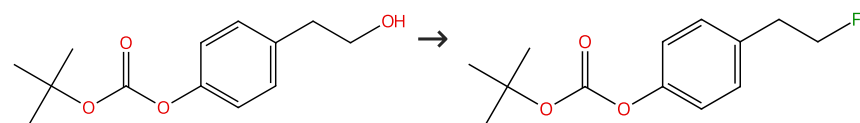
AlkylFluor: Deoxyfluorination of Alcohols

By: Goldberg, Nathaniel W.; et al

Organic Letters (2016), 18(23), 6102-6104.

Scheme 44 (1 Reaction)

Steps: 1 Yield: 98%



31-614-CAS-38381883

Steps: 1 Yield: 98%

1.1 Reagents: Triethylamine, Sulfur tetrafluoride
Solvents: Ethyl acetate; 5 min, 50 °C

1.2 Reagents: Sodium hydroxide
Solvents: Water

Experimental Protocols

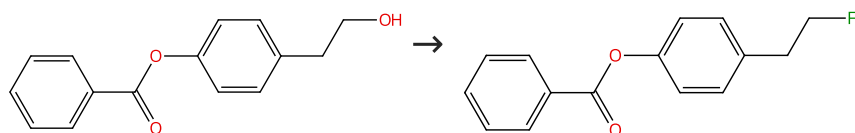
Sulfur Tetrafluoride (SF₄) as a Deoxyfluorination Reagent for Organic Synthesis in Continuous Flow Mode

By: Polterauer, Dominik; et al

Organic Process Research & Development (2023), 27(12), 2385-2392.

Scheme 45 (1 Reaction)

Steps: 1 Yield: 98%



Supplier (1)

31-614-CAS-38381878

Steps: 1 Yield: 98%

Sulfur Tetrafluoride (SF₄) as a Deoxyfluorination Reagent for Organic Synthesis in Continuous Flow Mode

By: Polterauer, Dominik; et al

Organic Process Research & Development (2023), 27(12), 2385-2392.

1.1 Reagents: Triethylamine, Sulfur tetrafluoride

Solvents: Ethyl acetate; 5 min, 50 °C

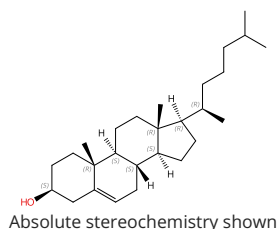
1.2 Reagents: Sodium hydroxide

Solvents: Water

Experimental Protocols

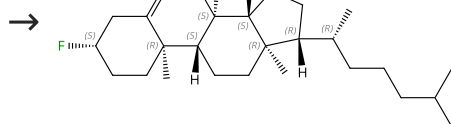
Scheme 46 (1 Reaction)

Steps: 1 Yield: 98%



Absolute stereochemistry shown

Suppliers (190)



Absolute stereochemistry shown

Suppliers (2)

31-048-CAS-14215707

Steps: 1 Yield: 98%

1,1,3,3,3-Pentafluoropropene secondary amine adducts as new selective fluorinating agents

By: Koroniak, Henryk; et al

Journal of Fluorine Chemistry (2006), 127(9), 1245-1251.

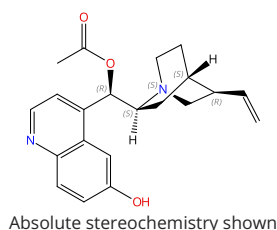
1.1 Reagents: 1-(1,3,3,3-Tetrafluoro-1-propen-1-yl)pyrrolidine, 1-

(1,1,3,3,3-Pentafluoropropyl)pyrrolidine

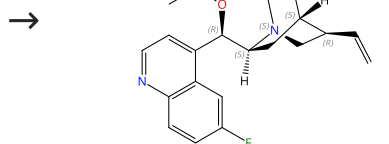
Solvents: Diethyl ether; 4 - 20 h, 60 °C

Scheme 47 (1 Reaction)

Steps: 1 Yield: 98%



Absolute stereochemistry shown



Absolute stereochemistry shown

31-048-CAS-14515622

Steps: 1 Yield: 98%

PhenoFluorMix: Practical Chemoselective Deoxyfluorination of Phenols

By: Fujimoto, Teppei; et al

Organic Letters (2015), 17(3), 544-547.

1.1 Reagents: 1*H*-imidazolium, 1,3-bis[2,6-bis(1-methylethyl)

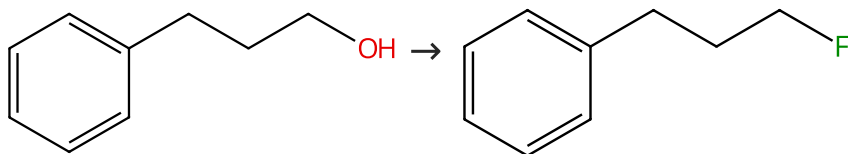
phenyl]-2-chloro-, chloride (1:1), mixt. with cesium fluoride

Solvents: Toluene; 30 min, 23 °C; 24 h, 110 °C

Experimental Protocols

Scheme 48 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (87)

Suppliers (10)

31-614-CAS-33631174

Steps: 1 Yield: 98%

Synthesis of Highly Reactive Sulfone Iminium Fluorides and Their Use in Deoxyfluorination and Sulfur Fluoride Exchange Chemistry

1.1 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene, 2815253-79-1
Solvents: Dichloromethane; 60 s, rt

Experimental Protocols

By: Vogel, James A.; et al

Organic Letters (2022), 24(32), 5962-5966.